Intelligent Cyber-attack Defense System using Virtual Honeypots for Cloud Security

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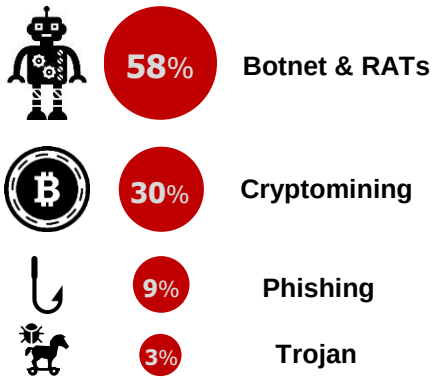
Defending against Cyber-attacks

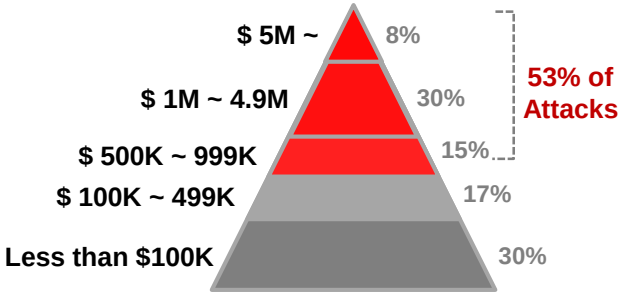
* Defenders should prepare to face new, sophisticated cyber threats

- ✓ Botnet activities appear to be rising to the top of security radar
- ✓ More than half of attacks are hardly detected or mitigated in advance



▲Cyber Attack Methods

▲Percentage of Investigated Attacks



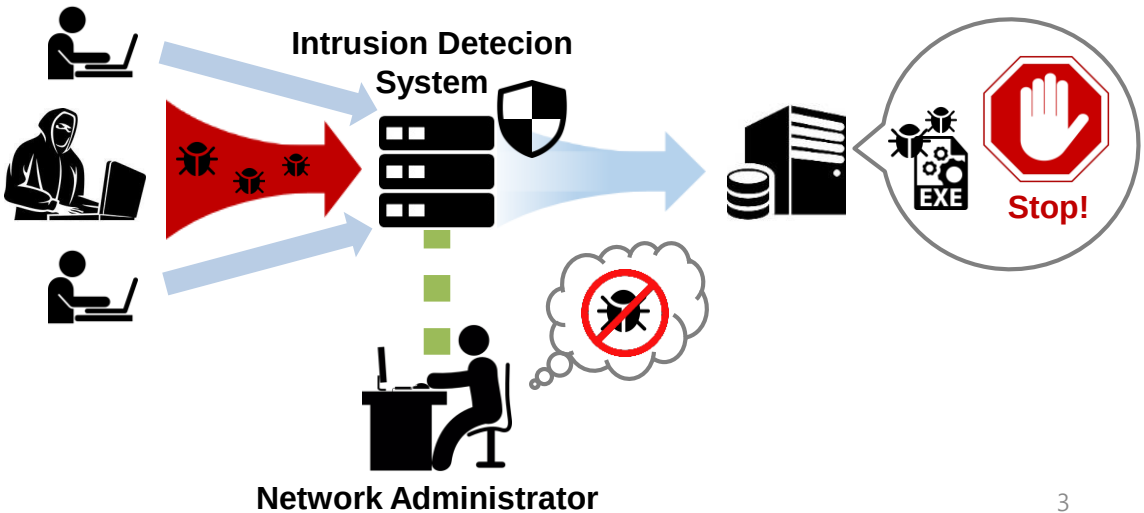
▲Financial Cost of Cyber Attacks

from 'Cisco annual cyber security reports 2018'

Need for Security Analytics Platforms

* **Security system requires advanced detection/protection technology :**

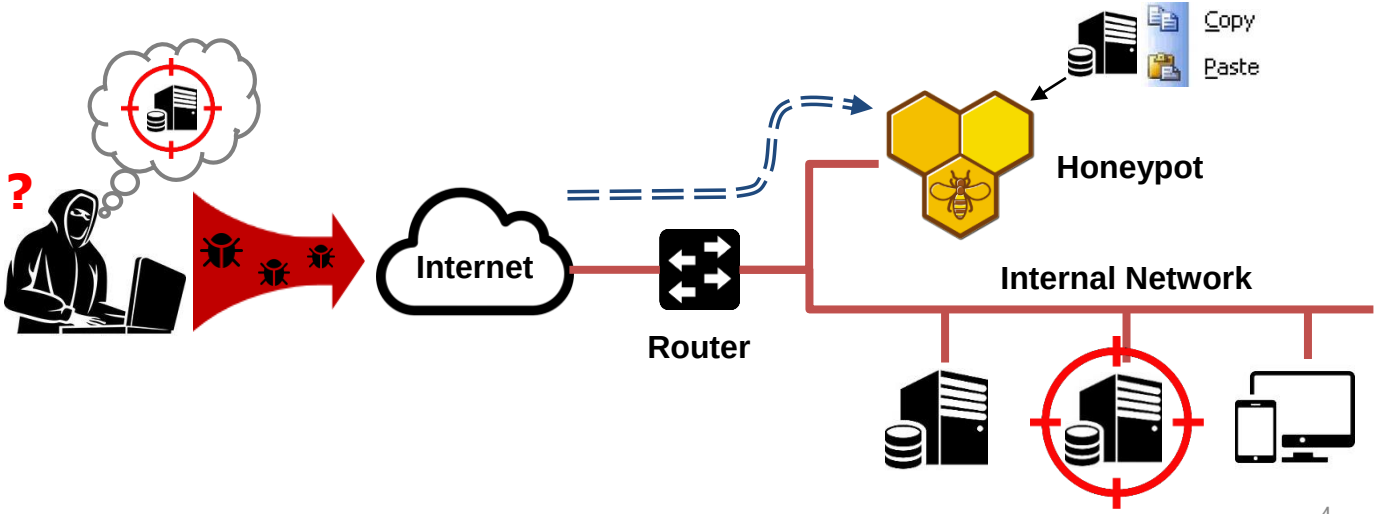
- ✓ Track unknown data flow
- ✓ Block the known malicious data flow
- ✓ Prevent the execution of attack campaigns



Honeypot System

* A security equipment intended to mimic likely target of cyber attacks

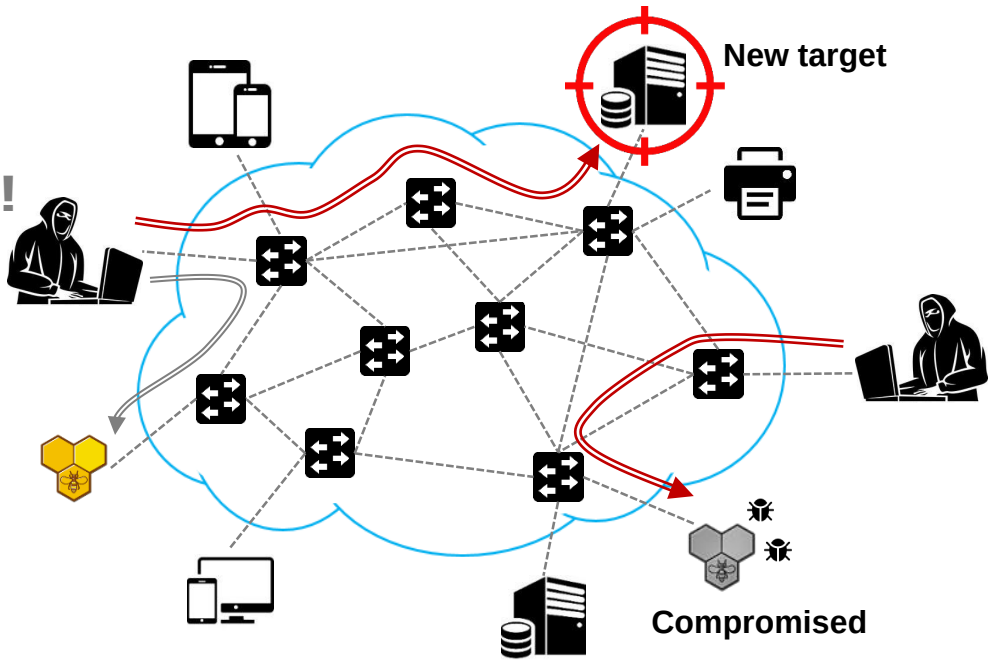
- ✓ Distract the attackers and prevent them from spreading damages to other networks
- ✓ Collect the information about the bad user, attack methods, tools



Challenges in Implementaion of Honeypot

* **Honeypot lack a ability for distinguishing the various attack scenarios**

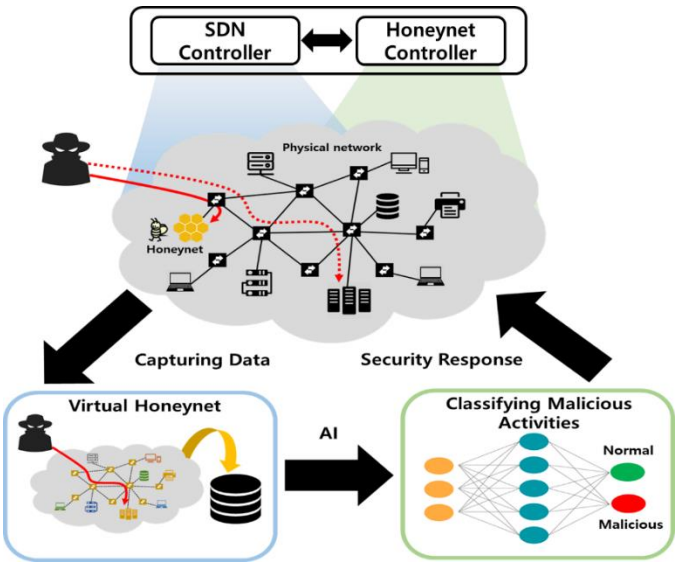
- ✓ Attackers can counter the existing measures to find out new target or they can exploit the honeypot system.



Main Idea

Proposed honeypot system

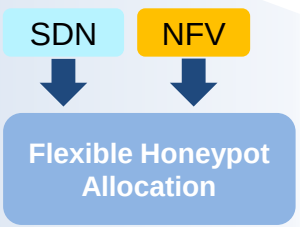
* Key Point : SDN/NFV management + AI based anomaly detection

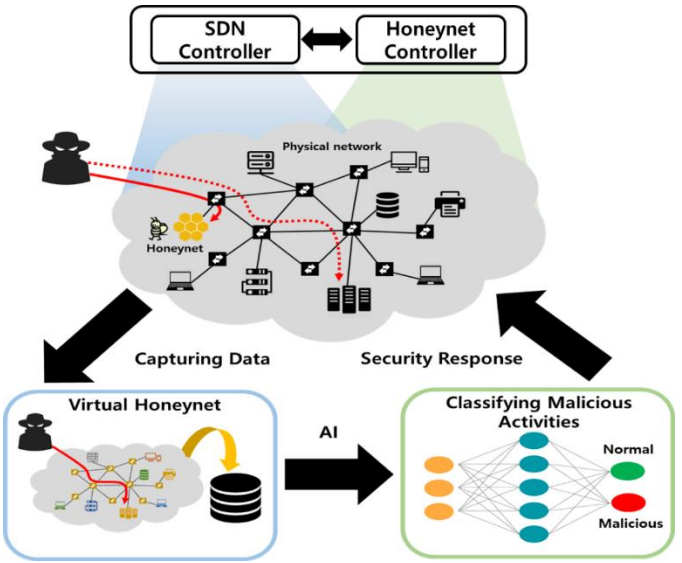


Proposed honeypot system

* Key Point : SDN/NFV management + AI based anomaly detection

- ✓ Software-Defined Network (SDN) → flexibility in controlling data flow
- ✓ Network Functions Virtualization (NFV) → faster honeypot deployment

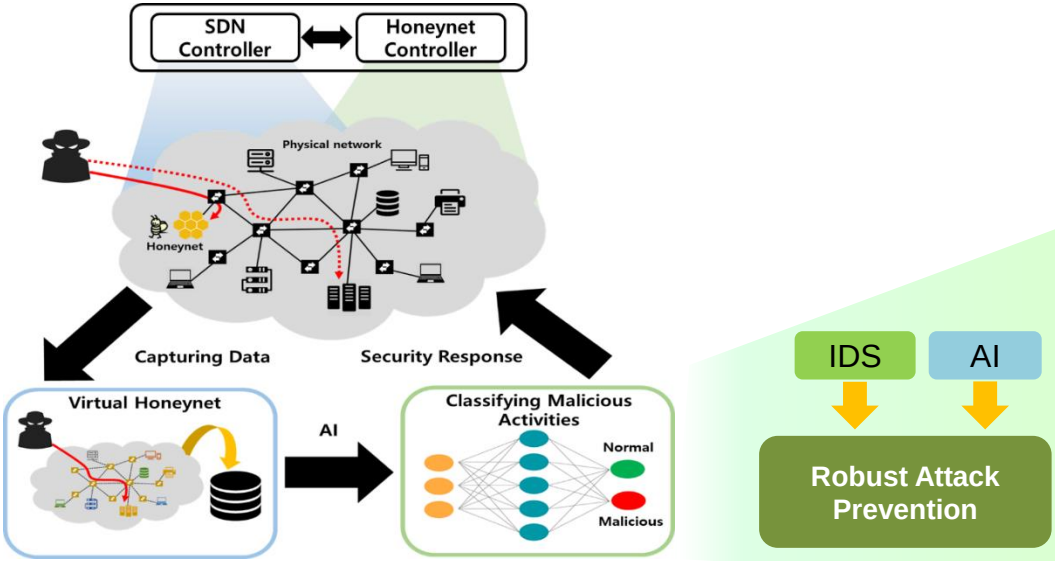




Proposed honeypot system

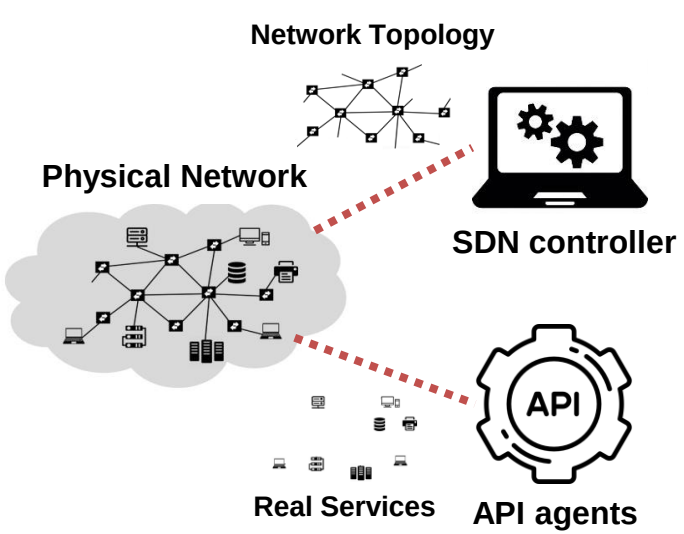
* Key Point : SDN/NFV management + AI based anomaly detection

- ✓ For determining and enforcing security guidelines, the controller are assisted by Deep Learning technology
- ✓ Traffic anomaly detection can potentially identify malicious activities



SDN/NFV Management

- ✓ Physical network consists of SDN enabled switches
- ✓ SDN controller obtains network topology while API agents collect the service information that the physical servers provide



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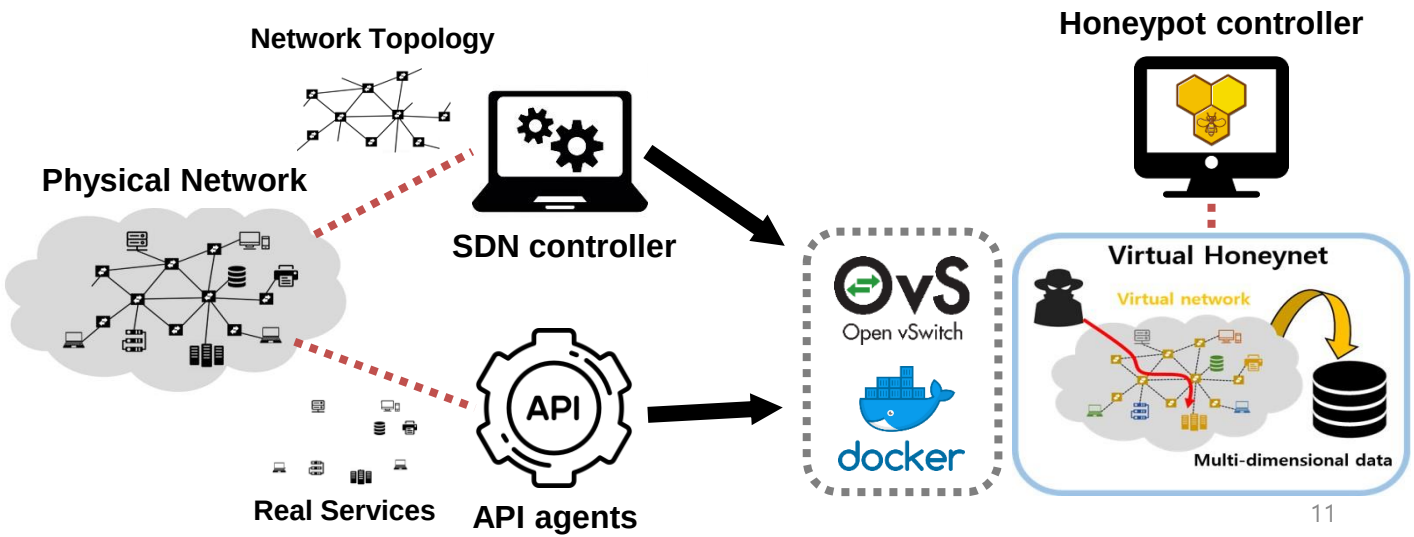
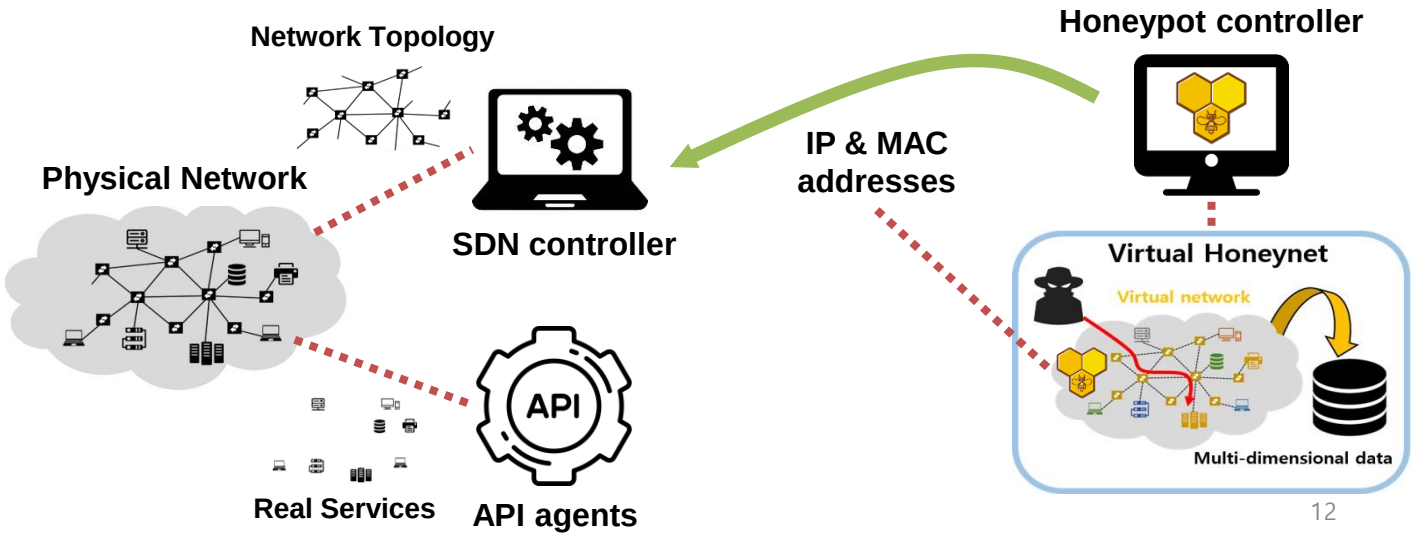
- ✓ Honeynet is built on Open vSwitch (OVS) and Docker images



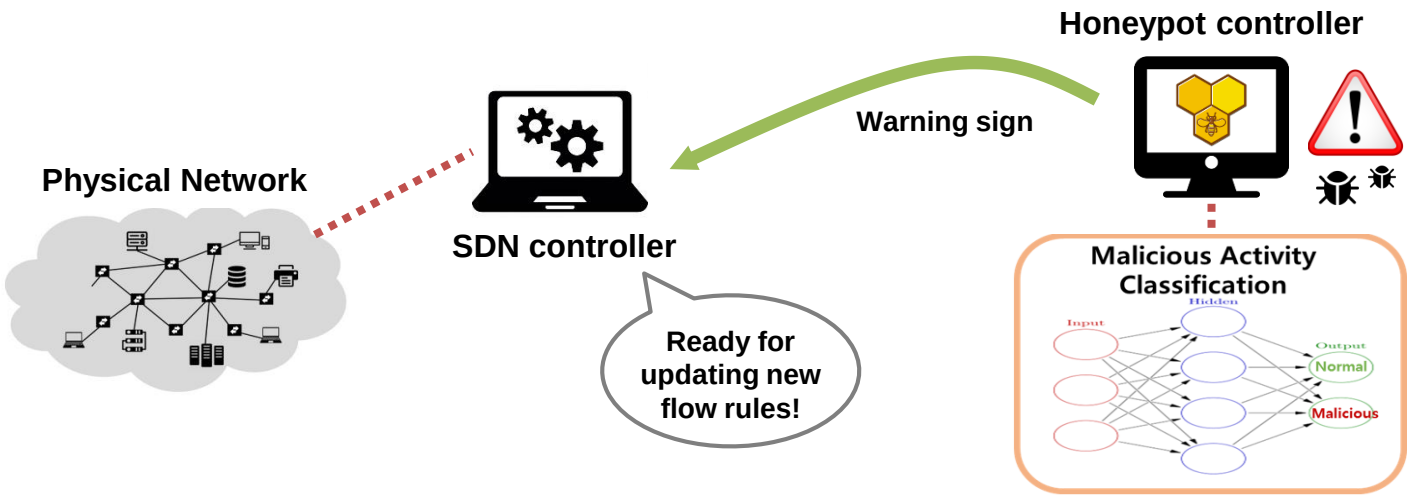
SDN/NFV Management

- ✓ Physical network consists of SDN enabled switches
- ✓ SDN controller obtains network topology while API agents collect the service information that the physical servers provide
- ✓ Honeynet is built on Open vSwitch (OVS), Openstack, and Docker images
- ✓ Honeypot controller replies with IP & MAC addresses of created honeypot



AI based anomaly detection

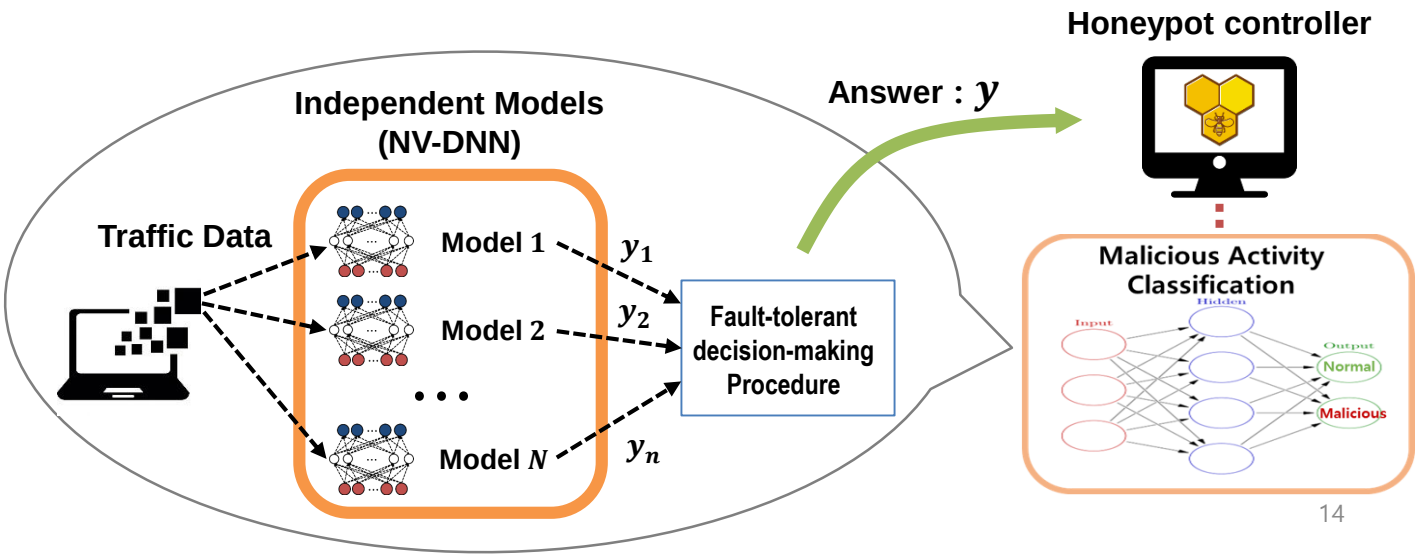
- ✓ Flow-level features are utilized to figure out the exploited channels
  - Average/Minimum/Maximum of TTL, TCP window, interarrival time, length of IP/TCP header, data payload for Client/Server side
  - Proportion of ACK/CWR/ECE/PSH/RST/SYN/URG
  - Percentage of duplicate ACK, keep alive, lost segments, retransmission, window update, zero window packets
- ✓ Honeypot controller sends a warning sign to the SDN controller to block the flow determined as abnormal



AI based anomaly detection

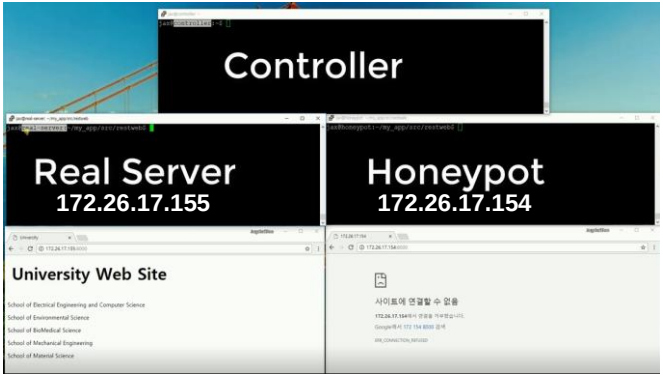
* **N-version Deep Neural Network (NV-DNN) : makes a decision with multiple simplex models**

- ✓ It is not quite easy to guarantee the reliability of a single model
- ✓ N-version programming (NVP) paradigm minimize the probability that independent models all produce erroneous results

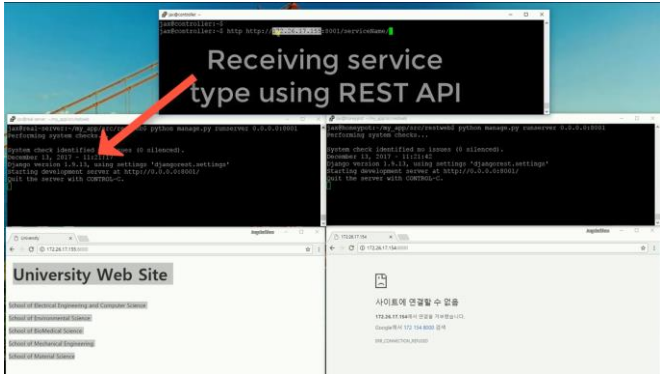


Experiment

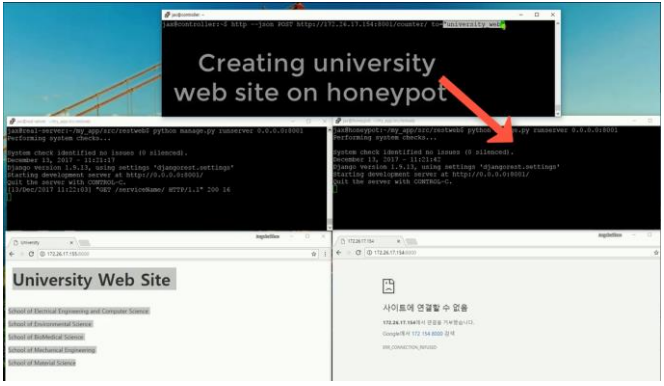
Honeypot creation



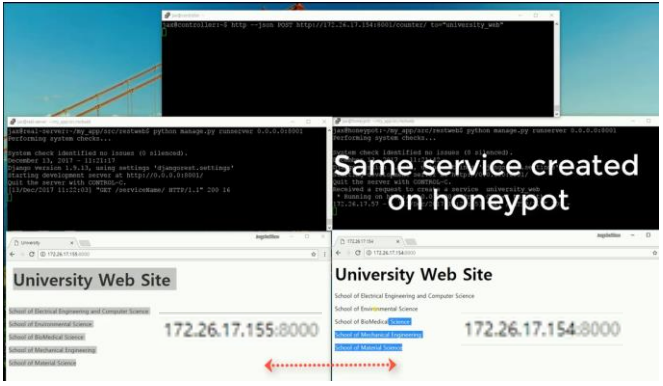
a) before collecting service information



b) receive service information



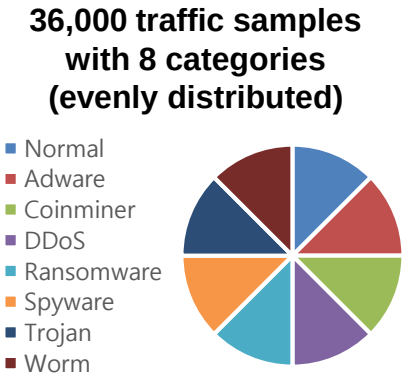
c) activate a service on honeypot



d) after running services on honeypot

Experiment

- ✓ dataset : 36,000 traffic samples with 8 categories (below chart)



Stratosphere Lab

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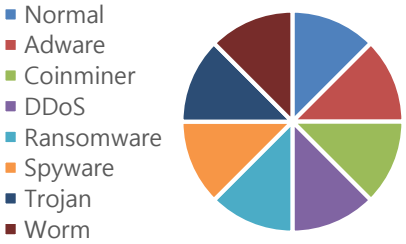
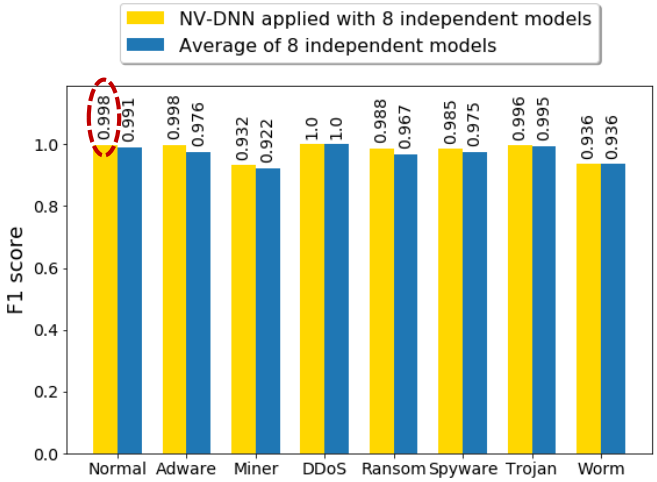
Experiment

- ✓ dataset : 36,000 traffic samples with 8 categories (below chart)
- ✓ Proposed classifier achieves 0.998 F1-score for normal category : high detection rate + low false alarm rate

36,000 traffic samples with 8 categories (evenly distributed)

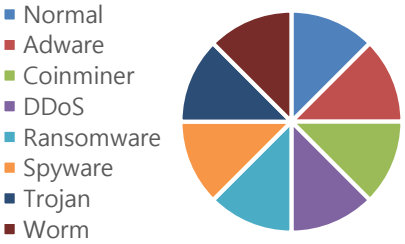


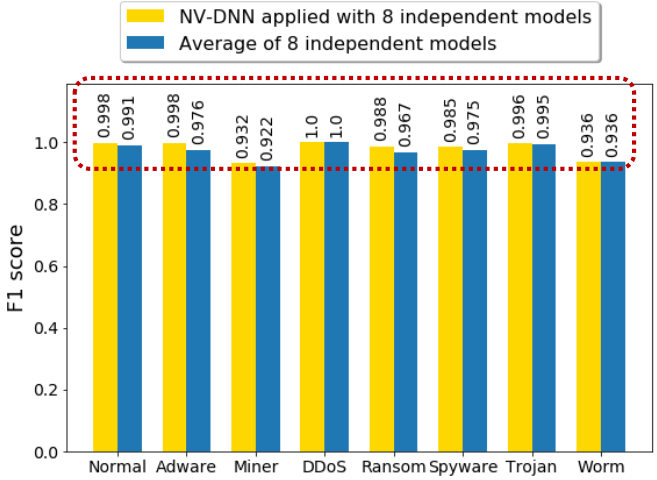


Experiment

- ✓ dataset : 36,000 traffic samples with 8 categories (below chart)
- ✓ Proposed classifier achieves 0.998 F1-score for normal category : high detection rate + low false alarm rate
- ✓ N-version approaches outperforms the independent models

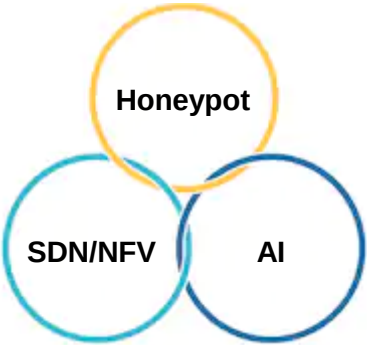
36,000 traffic samples with 8 categories (evenly distributed)



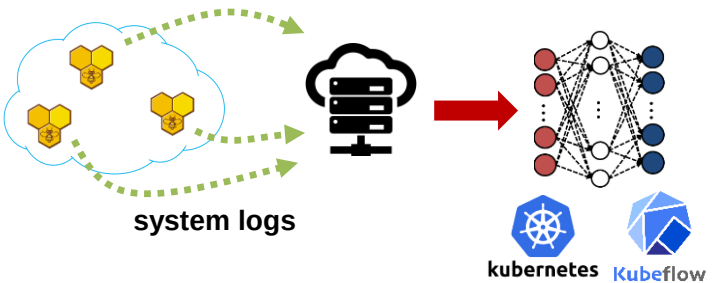


Conclusion & Future Works

- ✓ We designed an intelligent honeypot system with：
  - SDN/NFV management
  - AI-based traffic anomaly detection



- ✓ What we want to do next：
  - Analyze the pattern of honeypot response for anomaly detection
  - Automate the training to cope with constant change in attack patterns



Thank You.